Settlement Monitor: Anomaly Detection

What it does

I built a machine learning system that monitors simulated space settlement data. It tracks oxygen, pressure, temperature, CO2 and radiation, then detects abnormal events such as fires, hull breaches, solar flares and system failures.

The model

I trained a PPO reinforcement learning model to decide whether sensor readings are normal or indicate an anomaly. I tuned the reward system so that detecting an anomaly earlier gives a stronger reward, encouraging the model to react as quickly as possible while avoiding false alarms.

Why it is useful

Early detection could give people more time to respond to dangerous conditions in a space settlement. The simulation also allows different events, noise levels and severity levels to be tested safely.

Testing

The system tests the trained model and measures accuracy, precision, recall, false alarms and detection delay. It also shows the sensor data and where the model detected an event.

Completed

The project is complete. I have built the simulation, trained and tuned the PPO model, added automated testing and created tools to measure how accurately and quickly it detects anomalies.

Results

Model details

Model	settlement_20260922_210605
Trained at	2026-09-22 21:12:52
Trained type	Mars base
Trained params	[26.5, 70.0, 20.0, 1500.0, 12.0, 60.0, 0.6, 1.0, 1.0, 3.0]
PPO timesteps	300,000

Test episodes

Episode	Event	Start step	Caught after (steps)	False alarm steps
1	Scrubber failure	24	3	0
2	Solar flare	87	5	0
3	Solar flare	115	3	0
4	Fire	113	5	1
5	Fire	79	7	0
6	Hull breach	88	1	2
7	Heating failure	81	2	4
8	Hull breach	32	1	3
9	Heating failure	96	1	4
10	Hull breach	45	1	2

Overall performance

Metric	Value
True positives	100
False positives	16
Missed steps	41
True negatives	1343
Accuracy	0.962
Precision	0.862
Recall	0.709
Events caught	10 / 10
Average delay	2.9 steps

Event type	Caught
Hull breach	3 / 3
Scrubber failure	1 / 1
Solar flare	2 / 2
Heating failure	2 / 2
Fire	2 / 2

Sensor plots

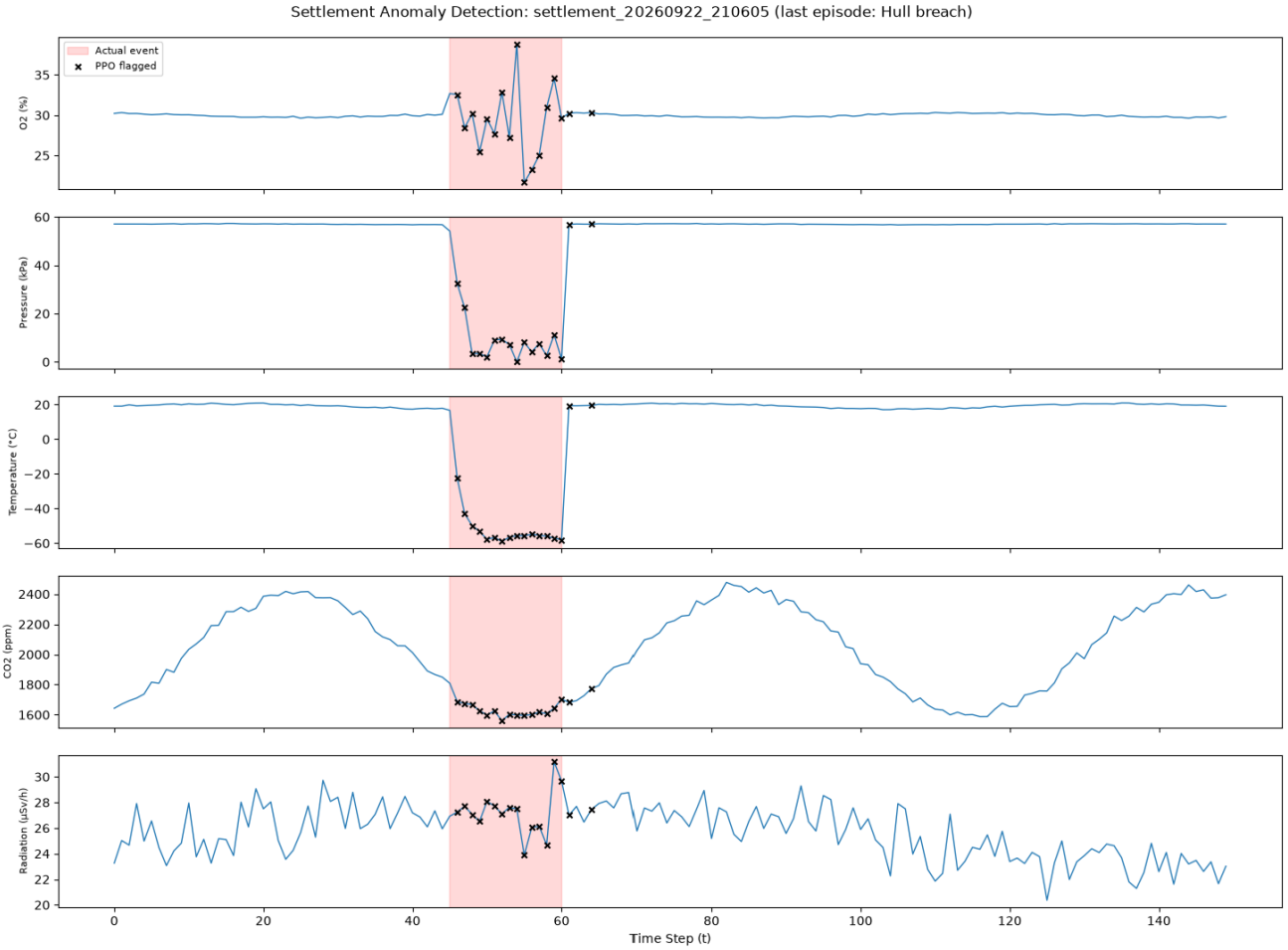


Figure 1: Model settlement_20260922_210605 on a hull breach starting at step 45. The shaded area is the actual event and the crosses are the steps the PPO model flagged.

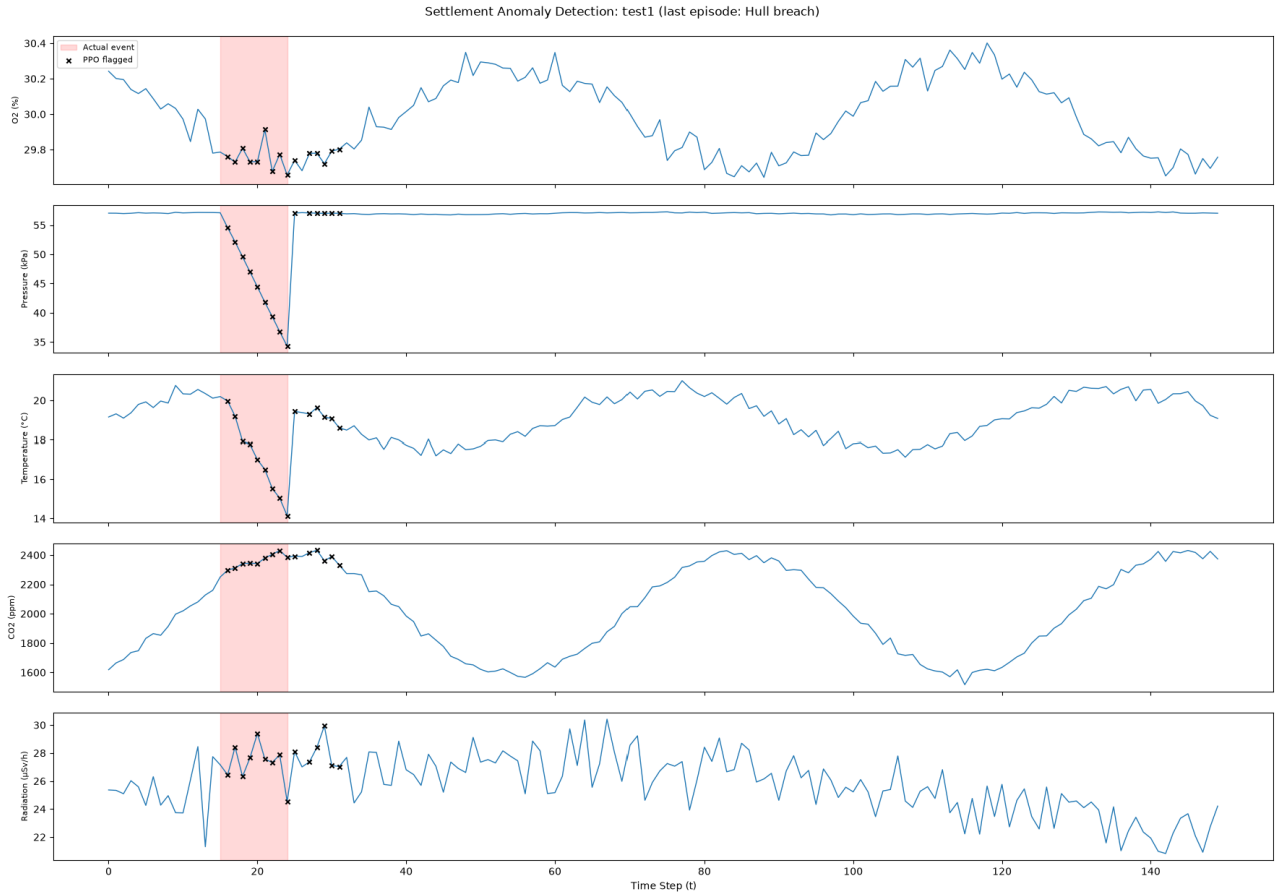


Figure 2: Test run test1 on a hull breach starting at step 15, showing the model tracking a slower pressure and temperature drop.